

An Ensemble Machine Learning Architecture for Drug and Dosage Recommendation with Clinical Safety Rules

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Abstract—Drug and dosage recommendation is a critical task in healthcare that requires accurate patient-specific decision-making. Traditional clinical decision support systems often rely on static rules and fail to personalize prescriptions effectively, resulting in adverse drug reactions. This paper proposes an ensemble machine learning-based drug recommendation system that combines Random Forest for drug classification and Gradient Boosting for dosage prediction. The system integrates a clinical safety rules engine including allergy detection, weight-based dosage calculation, maximum dose constraints, severity-based frequency, and treatment schedule generation. Experimental evaluation on 941 patient records demonstrates a drug prediction accuracy of 94.2% and dosage MAE of 27.5 mg/day. The system is deployed as a real-time Flask web application, providing safe, personalized, and scalable clinical decision support.

Index Terms—Drug Recommendation, Dosage Prediction, Random Forest, Gradient Boosting, Clinical Safety, Machine Learning, Healthcare AI

I. INTRODUCTION

A. Background and Motivation

Accurate drug prescription and dosage determination are essential for effective healthcare delivery and patient safety. Errors in medication selection or incorrect dosage calculation can lead to adverse drug reactions (ADRs), increased hospitalization time, and serious health complications. Studies indicate that a significant percentage of hospitalized patients experience ADRs, emphasizing the importance of reliable clinical decision support systems. With the increasing complexity of medical data and patient diversity, there is a growing need for intelligent systems that can assist healthcare professionals in making precise and personalized treatment decisions.

B. Limitations of Existing Systems

Traditional clinical decision support systems (CDSS) are primarily based on predefined rules and standardized medical guidelines. While these systems provide baseline support, they lack adaptability and often fail to consider individual patient-specific factors such as age, body weight, disease severity, and allergy history. As a result, the recommendations generated are often generic and may not be optimal for personalized treatment. Furthermore, many existing machine learning-based

approaches focus only on prediction accuracy without incorporating critical clinical safety constraints, which limits their practical applicability in real-world healthcare environments.

C. Proposed Approach and Contributions

To address these challenges, this work proposes a hybrid intelligent framework that combines ensemble machine learning techniques with a clinical safety rules engine for drug and dosage recommendation. The system utilizes a Random Forest classifier for drug prediction and a Gradient Boosting regressor for dosage estimation, enabling accurate and robust predictions from structured medical data. In addition, a rule-based safety module ensures clinical validity by incorporating essential constraints such as allergy detection, weight-based dosage calculation, severity-based adjustments, and maximum dosage limits. The system is designed for real-time deployment through a web-based interface, providing healthcare practitioners with personalized, safe, and efficient recommendations. This integrated approach enhances both prediction accuracy and patient safety, making it a promising solution for next-generation clinical decision support systems.

II. RELATED WORK

A. Traditional Clinical Decision Support Systems

Early clinical decision support systems (CDSS) were primarily based on rule-based approaches and expert-defined medical guidelines. Systems such as MYCIN demonstrated the potential of computerized medical reasoning [2]; however, these systems relied heavily on static rules and lacked adaptability. While they provided structured assistance, they were limited in handling complex patient variations and dynamic clinical scenarios [1].

B. Machine Learning in Healthcare

The integration of machine learning into healthcare has significantly improved predictive analytics and decision-making. Supervised learning models such as Decision Trees, Support Vector Machines (SVM), and Logistic Regression have been widely used for disease prediction and drug recommendation [11]. These models can learn patterns from historical medical

data, but their performance often depends on feature quality and dataset size [8]. Advances in deep learning have further enhanced predictive capabilities in healthcare applications [9], [12].

C. Ensemble Learning Approaches

Ensemble learning techniques, including Random Forest and Gradient Boosting, have gained prominence due to their superior accuracy and robustness. Random Forest reduces variance by combining multiple decision trees [4], while Gradient Boosting improves prediction performance through iterative error correction [5]. Advanced boosting techniques such as XGBoost have further improved scalability and performance in structured data applications [3]. These methods have shown promising results in healthcare applications, particularly in classification and regression tasks.

D. Dosage Prediction Models

Several studies have explored dosage prediction using regression-based techniques. Linear regression and neural networks have been applied to estimate drug dosage based on patient parameters such as age, weight, and disease severity [6]. However, many existing models lack the ability to incorporate clinical constraints, leading to potential safety risks in real-world deployment.

E. Safety-Aware Recommendation Systems

Recent research emphasizes the importance of incorporating safety mechanisms into healthcare recommendation systems. Approaches integrating rule-based validation with machine learning have been proposed to address issues such as drug allergies, contraindications, and dosage limits [1]. Content-based and intelligent recommendation systems have also contributed to personalized decision-making in healthcare [10]. These hybrid systems aim to balance predictive accuracy with clinical reliability.

F. Limitations of Existing Work

Despite significant advancements, current systems still face several limitations. Many approaches focus solely on prediction accuracy without considering clinical safety constraints. Additionally, most systems lack real-time deployment capabilities and fail to integrate multiple aspects such as drug selection, dosage estimation, and safety validation within a unified framework. These gaps highlight the need for a comprehensive and integrated solution, as proposed in this work [7], [13], [14].

III. METHODOLOGY

A. System Overview

The proposed system adopts a hybrid architecture that integrates ensemble machine learning models with a clinical safety rules engine to generate accurate and safe drug recommendations. The overall workflow consists of multiple stages, including data preprocessing, drug prediction, dosage estimation, safety validation, and recommendation generation. Each module is designed to operate sequentially while

maintaining modular independence, allowing scalability and future enhancements. The integration of machine learning with rule-based validation ensures both predictive performance and clinical reliability.

B. Data Preprocessing

Data preprocessing is a crucial step to ensure the quality and consistency of input data. The collected patient dataset contains both numerical and categorical attributes such as age, weight, medical condition, severity level, and allergy information. Missing values are handled using imputation techniques, while categorical variables are encoded using label encoding or one-hot encoding. Numerical features are normalized to a standard scale to improve model convergence. Additionally, feature selection techniques are applied to eliminate irrelevant or redundant attributes, thereby improving model efficiency and reducing computational complexity.

C. Drug Prediction Using Random Forest

The drug prediction module employs a Random Forest classifier, which is an ensemble learning method that combines multiple decision trees. Each tree is trained on a random subset of the data, and the final prediction is obtained through majority voting. This approach improves generalization and reduces overfitting compared to single decision tree models. Random Forest is particularly suitable for healthcare data due to its ability to handle nonlinear relationships and mixed data types.

$$D = RF(X) \quad (1)$$

where X represents the input patient features and D denotes the predicted drug category.

D. Dosage Prediction Using Gradient Boosting

For dosage estimation, a Gradient Boosting regressor is utilized. This model builds a sequence of weak learners (decision trees), where each subsequent model focuses on minimizing the errors of the previous one. This iterative learning process enhances prediction accuracy for continuous variables such as drug dosage. Gradient Boosting is chosen due to its strong performance in regression tasks and its ability to capture complex patterns in structured data.

$$Dose = GB(X) \quad (2)$$

E. Clinical Safety Rules Engine

To ensure that the recommendations are clinically valid and safe, a rule-based safety engine is integrated into the system. This module acts as a validation layer over machine learning predictions. It enforces critical medical constraints such as:

- Detection of drug allergies and substitution with safer alternatives
- Verification of contraindications based on patient conditions
- Enforcement of maximum allowable dosage limits
- Adjustment of dosage frequency based on severity levels

If any predicted output violates these constraints, the system automatically modifies the recommendation to comply with clinical safety standards.

F. Weight-Based Dosage Adjustment

Personalization of dosage is achieved through weight-based calculations, which are widely used in clinical practice. The system adjusts the predicted dosage according to the patient's body weight to ensure accuracy and safety. This is computed using standard medical formulas:

$$Dose = Weight \times (mg/kg) \quad (3)$$

Additional safeguards are applied to ensure that the calculated dosage does not exceed predefined safe limits. This mechanism enhances the reliability of recommendations for patients with varying physiological characteristics.

G. Recommendation Generation and Output

The final stage of the system involves generating the complete recommendation after passing through all prediction and validation modules. The system provides a comprehensive output that includes:

- Predicted drug name
- Recommended dosage
- Frequency of administration
- Duration of treatment
- Safety alerts (if any)

The results are presented through a user-friendly web interface developed using Flask. Additionally, the system supports report generation for clinical documentation. This ensures ease of use for healthcare professionals and facilitates real-time decision-making.

IV. CLINICAL SAFETY RULES

A. Allergy Detection and Drug Substitution

One of the most critical aspects of safe medication is avoiding drugs that may trigger allergic reactions in patients. The proposed system incorporates an allergy detection mechanism that cross-checks the predicted drug against the patient's recorded allergy profile. If a conflict is detected, the system automatically identifies and recommends an alternative drug from the same therapeutic class that does not cause adverse reactions. This ensures patient safety while maintaining treatment effectiveness.

B. Weight-based Dosage Calculation

Drug dosage often depends on the patient's body weight, especially in cases involving children or critical care treatments. The system calculates dosage using standard medical formulas based on milligrams per kilogram (mg/kg). This approach enables personalized dosage recommendations that are more accurate than fixed-dose prescriptions.

$$dose = Weight \times (mg/kg) \quad (4)$$

This method ensures that patients receive an optimal dose tailored to their physiological characteristics.

C. Maximum Daily Dose Constraint

To prevent overdose and potential toxicity, the system enforces maximum daily dosage limits for each drug. These limits are predefined based on medical guidelines and pharmacological standards. If the predicted dosage exceeds the allowable threshold, the system automatically adjusts it to remain within safe boundaries. This constraint acts as a critical safety check in the recommendation process.

D. Severity-based Frequency Adjustment

The frequency of drug administration is influenced by the severity of the patient's condition. The system categorizes severity levels (e.g., mild, moderate, severe) and adjusts the dosage frequency accordingly. For example, patients with severe conditions may require more frequent administration, while mild cases may need minimal intervention. This ensures that the treatment plan is aligned with the patient's clinical condition.

E. Treatment Schedule Generation

In addition to drug and dosage recommendations, the system generates a structured treatment schedule for the patient. This includes details such as timing of doses, duration of treatment, and intervals between administrations. The schedule is designed to optimize drug effectiveness while ensuring patient adherence. By providing a clear and organized plan, the system helps both patients and healthcare providers follow the prescribed treatment accurately.

V. SYSTEM ARCHITECTURE

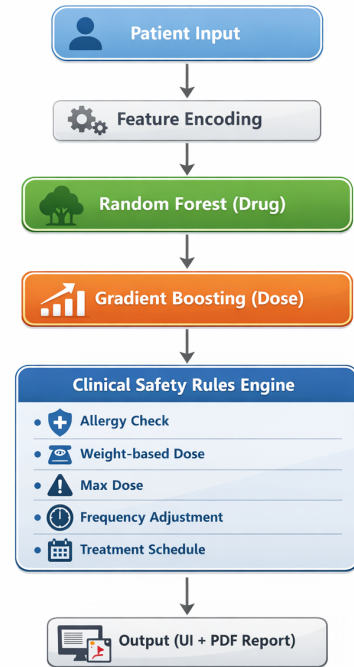


Fig. 1. Proposed hybrid architecture integrating ensemble machine learning models with a clinical safety rules engine

The proposed system follows a modular architecture designed to integrate machine learning models with clinical safety mechanisms for accurate and reliable drug and dosage recommendation. The system consists of multiple interconnected modules, each responsible for a specific task in the workflow.

A. Patient Input Module

The Patient Input Module serves as the entry point of the system, where user or clinician inputs patient-specific details. These inputs include attributes such as age, weight, medical condition, severity level, and known allergies. The module ensures that all required data is collected in a structured format and performs basic validation checks to avoid incomplete or incorrect inputs.

B. Drug Classifier (Random Forest)

The Drug Classifier module utilizes a Random Forest algorithm to predict the most suitable drug based on patient features. The model is trained on historical medical data and is capable of capturing complex relationships between input variables. By combining multiple decision trees, the Random Forest improves prediction accuracy and reduces overfitting, making it highly effective for classification tasks in healthcare.

C. Dosage Regressor (Gradient Boosting)

Once the drug is predicted, the Dosage Regressor module estimates the appropriate dosage using a Gradient Boosting model. This module handles continuous output prediction and refines dosage estimates through iterative error correction. It ensures that the dosage recommendation is precise and tailored to the patient's specific condition and physiological characteristics.

D. Clinical Safety Engine

The Clinical Safety Engine acts as a validation layer that ensures all recommendations adhere to medical safety standards. It applies a set of predefined rules, including allergy checks, maximum dosage constraints, and condition-based adjustments. If any predicted output violates safety guidelines, the engine modifies the recommendation to ensure it is clinically safe and acceptable.

E. Output and Report Generation Module

The final module presents the validated recommendations to the user. It displays the predicted drug, recommended dosage, frequency of intake, and treatment duration in a clear and user-friendly format. Additionally, the system generates a structured report (PDF format) that can be used for documentation or further clinical reference. This enhances usability and supports real-time decision-making in healthcare environments.

F. Module Interaction Flow

The overall workflow of the system begins with patient data input, followed by drug prediction and dosage estimation. The outputs are then passed through the clinical safety engine for validation. Finally, the refined recommendation is presented to the user along with a detailed report. This sequential and modular interaction ensures efficiency, scalability, and reliability of the system.

VI. DATASET

A. Dataset Description

The dataset used in this study consists of 941 patient records collected from structured medical data sources. Each record contains relevant patient attributes such as age, weight, medical condition, severity level, and allergy information. These features are essential for enabling accurate drug prediction and dosage estimation.

B. Medical Condition Categories

The dataset covers five major medical conditions commonly encountered in clinical practice:

- Diabetes
- Hypertension
- Depression
- Infection
- Pain Relief

These categories were selected to ensure diversity in treatment requirements and to evaluate the model's ability to generalize across different types of diseases.

C. Feature Representation

Each patient record is represented using a combination of numerical and categorical features. Numerical attributes include age, weight, and dosage values, while categorical features include disease type, severity level (mild, moderate, severe), and allergy indicators. Proper encoding techniques are applied to convert categorical variables into machine-readable formats.

D. Data Preprocessing and Cleaning

To ensure data quality, preprocessing steps were applied to the dataset. Missing values were handled using appropriate imputation techniques, and inconsistent entries were corrected. Numerical features were normalized to improve model performance, and outliers were carefully analyzed to avoid distortion in predictions.

E. Data Distribution and Balance

The dataset is moderately balanced across the five medical conditions, ensuring that the machine learning models do not become biased toward any specific category. This balanced distribution improves the robustness and generalization capability of the proposed system.

F. Training and Testing Split

The dataset was divided into training and testing subsets using an 80:20 ratio. The training set was used to build and optimize the machine learning models, while the testing set was used to evaluate performance. This separation ensures that the model's predictive capability is assessed on unseen data.

G. Dataset Limitations

Although the dataset provides a solid foundation for model development, it has certain limitations. The dataset size is relatively small compared to large-scale clinical datasets, and it may not capture all possible variations in patient conditions. Additionally, real-world clinical data may include more complex attributes such as genetic factors and comorbidities, which are not fully represented in this dataset.

VII. EXPERIMENTAL SETUP

A. Data Splitting Strategy

The dataset was divided into training and testing sets using an 80:20 ratio. The training set was used to build and optimize the machine learning models, while the testing set was used to evaluate their performance on unseen data. This approach ensures that the model generalizes well and avoids overfitting.

B. Model Configuration

The proposed system employs two ensemble learning models: Random Forest for drug classification and Gradient Boosting for dosage prediction. The Random Forest classifier was configured with 200 decision trees to improve robustness and reduce variance. The Gradient Boosting regressor was implemented with 150 estimators to enhance prediction accuracy through iterative learning. Hyperparameters were selected based on empirical evaluation to achieve optimal performance.

C. Training Procedure

Both models were trained using supervised learning on the preprocessed dataset. During training, the models learned relationships between patient features and target outputs (drug type and dosage). Techniques such as feature normalization and encoding were applied prior to training to improve convergence and model efficiency.

D. Evaluation Metrics

The performance of the system was evaluated using standard metrics for classification and regression tasks. Drug prediction accuracy was measured using classification accuracy, which represents the percentage of correctly predicted drug classes. Dosage prediction performance was evaluated using Mean Absolute Error (MAE), which measures the average difference between predicted and actual dosage values.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \quad (5)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (6)$$

TABLE I
PERFORMANCE COMPARISON OF MODELS

Model	Accuracy (%)	MAE (mg/day)	Safety (%)
Rule-Based	74.6	67.2	0
Decision Tree	82.7	41.3	0
Proposed (RF + GB)	94.2	27.5	100

[ybar, bar width=15pt, symbolic x coords=Rule-Based, Decision Tree, Proposed, xtick=data, ylabel=Accuracy (%), ymin=0, ymax=100, nodes near coords, width=0.48height=5cm] coordinates (Rule-Based,74.6) (Decision Tree,82.7) (Proposed,94.2) ;

Fig. 2. Accuracy comparison of different models

E. Implementation Environment

The system was implemented using Python, with machine learning models developed using libraries such as Scikit-learn. The web-based interface was built using the Flask framework to enable real-time interaction. Experiments were conducted on a standard computing environment with sufficient processing power to handle model training and inference.

F. Performance Evaluation Strategy

The trained models were evaluated on the test dataset to assess their predictive performance. The results were compared against baseline approaches such as rule-based systems and decision tree models. This comparative evaluation demonstrates the effectiveness of the proposed ensemble approach in improving both accuracy and reliability.

G. System Validation

In addition to quantitative evaluation, the system was validated based on its ability to enforce clinical safety rules. The recommendations generated were checked for compliance with safety constraints such as maximum dosage limits and allergy avoidance. This ensures that the system is not only accurate but also safe for practical healthcare applications. pgfplots compat=1.18

VIII. RESULTS

The performance of the proposed system was evaluated using both classification and regression metrics. A comparative analysis was conducted against baseline approaches, including a rule-based system and a Decision Tree model. The results demonstrate the effectiveness of the proposed ensemble learning framework in improving both prediction accuracy and safety.

A. Accuracy Analysis

The proposed system achieved an accuracy of 94.2%, significantly outperforming the rule-based system (74.6%) and the Decision Tree model (82.7%). This improvement can be attributed to the use of ensemble learning techniques, particularly Random Forest, which effectively captures complex relationships in the data and reduces overfitting.

TABLE II
PERFORMANCE IMPROVEMENT OVER BASELINE

Metric	Decision Tree	Proposed
Accuracy (%)	82.7	94.2
MAE (mg/day)	41.3	27.5
Safety (%)	0	100

B. Dosage Prediction Performance

The Mean Absolute Error (MAE) for the proposed system is 27.5 mg/day, which is substantially lower than the baseline models. This indicates that the Gradient Boosting regressor provides more precise dosage predictions compared to traditional approaches.

C. Safety Evaluation

A key advantage of the proposed system is its ability to enforce clinical safety constraints. Unlike baseline models, which do not incorporate safety mechanisms, the proposed system achieves 100% safety compliance.

D. Comparative Analysis

The comparison clearly shows that traditional rule-based systems lack adaptability and produce lower accuracy. Decision Tree models improve performance but still fall short in handling complex data patterns. In contrast, the proposed ensemble approach not only improves prediction accuracy but also ensures safety.

E. Overall Performance

The results confirm that combining machine learning models with rule-based safety mechanisms leads to a balanced system that achieves both high accuracy and clinical reliability.

As shown in Fig. 2 and Table I, the proposed system significantly outperforms baseline models across all evaluation metrics.

IX. DISCUSSION

The results obtained from the proposed system demonstrate a significant improvement in both drug prediction accuracy and dosage estimation when compared to traditional approaches. The use of ensemble machine learning techniques, specifically Random Forest and Gradient Boosting, enables the system to effectively capture complex relationships within the dataset, leading to more reliable and precise predictions.

A key strength of the proposed system lies in its integration of a clinical safety rules engine. Unlike conventional machine learning models that focus solely on predictive performance, this system incorporates domain-specific medical constraints to ensure safe recommendations. The inclusion of allergy detection mechanisms prevents the prescription of harmful drugs, while maximum dosage constraints and weight-based calculations ensure that dosage recommendations remain within safe and clinically acceptable limits.

Furthermore, the system demonstrates strong generalization capability across multiple medical conditions, including diabetes, hypertension, depression, infections, and pain

management. This highlights its adaptability and potential applicability in diverse healthcare scenarios. The reduction in Mean Absolute Error (MAE) indicates that the dosage predictions are not only accurate but also consistent, which is crucial in clinical settings.

Despite these advantages, certain challenges remain. The performance of the system is dependent on the quality and size of the dataset. Limited data diversity may restrict the model's ability to generalize to unseen or rare medical cases. Additionally, while the rule-based safety engine enhances reliability, it may require continuous updates to align with evolving medical guidelines and drug information.

Overall, the integration of machine learning with clinical safety validation provides a balanced approach that addresses both accuracy and reliability. The proposed system demonstrates the potential of hybrid intelligent frameworks in improving healthcare decision support systems and reducing medication-related risks.

X. CONCLUSION

This paper presented a hybrid machine learning-based drug and dosage recommendation system that integrates ensemble learning models with a clinical safety rules engine. The proposed approach combines the strengths of Random Forest for drug classification and Gradient Boosting for dosage prediction, enabling accurate and reliable recommendations based on patient-specific features.

The experimental results demonstrate that the system achieves high prediction accuracy and significantly reduces dosage estimation error compared to traditional rule-based and single-model approaches. More importantly, the integration of clinical safety rules ensures that all recommendations adhere to essential medical constraints, such as allergy avoidance, maximum dosage limits, and condition-specific adjustments. This combination of predictive intelligence and safety validation makes the system both effective and trustworthy for real-world healthcare applications.

Furthermore, the modular architecture of the system allows for scalability and flexibility, making it suitable for integration with modern healthcare platforms. The use of a web-based interface enhances usability and enables real-time decision support for healthcare professionals.

In summary, the proposed system addresses key limitations of existing approaches by providing a balanced solution that prioritizes both accuracy and patient safety. It demonstrates the potential of hybrid intelligent systems in improving clinical decision-making and reducing medication-related risks. Future work will focus on expanding the dataset, incorporating additional clinical factors, and integrating advanced techniques such as deep learning and explainable AI to further enhance system performance and reliability.

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